Reciprocating the log term of equation 1.5

$$pH = pK_a + \log_{10} \frac{[A^-]}{[HA]} \quad \text{.....(1.6)}$$

This form of the ionization constant equation is called the Henderson-Hasselbalch equation (equation 1.6). It is useful for calculating the pH of a weak acid solution containing its conjugate base (salt). The other forms of the Henderson-Hasselbalch equation of weak acid with its conjugate base are as follows:

$$pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{Acid}]} \quad \text{.....(1.7)}$$

or

$$pH = pK_a + \log \frac{[\text{proton aceptor}]}{[\text{proton donor}]}$$

When the concentrations of weak acid and its conjugate base or weak base and its conjugate acid are equal, the pH of the solution equals the pK_a of the buffer. This is evident from the Henderson-Hasselbalch equation.

If the pK_a of bicarbonate buffer is 6.1 and the pH of blood is 7.4, then the ratio of bicarbonate to carbonic acid ($[HCO_3^-]$ / $[H_2CO_3]$) in blood is calculated by applying the Henderson-Hasselbalch equation.

$$pH = pK_a + \log \frac{[HCO_3^-]}{[H_2CO_3]} \quad \text{..... (1.8)}$$

Substitute the pH and pK_a values into the Henderson–Hasselbalch equation.

$$7.4 = 6.1 + \log \frac{[HCO_3^-]}{[H_2CO_3]} \quad \text{.....(1.9)}$$

Rearrange the equation 1.9

$$\log \frac{[HCO_3^-]}{[H_2CO_3]} = 1.3 \quad \text{.....(1.10)}$$

Changing log term to the other side of the equation 1.10

$$\frac{[HCO_3^-]}{[H_2CO_3]} = \text{antilog } 1.3 \quad \text{.....(1.11)}$$

or

$$\frac{[HCO_3^-]}{[H_2CO_3]} = 20 \quad \text{.....(1.12)}$$

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